An Alignment Check for Debiasing Corrections: A Negative-Control Case Study in Drug Repurposing

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Abstract. A bias-diagnostic metric can judge a debiasing correction only if it is sensitive to what that correction changes; two corrections with different mechanisms can target different quantities, and a metric tuned to one will misjudge the other regardless of which is substantively better. We formalize this as an alignment check: score each correction's own penalty term as a standalone classifier of whatever the bias metric flags, treating near-chance AUROC as a screening signal, not proof, that the metric cannot see that mechanism, and confirm or refute it with a direct intervention test. We develop the check through a case study in drug repurposing, where a model trained to rank candidates for one indication also learns what is shared across drugs on the same organ system; in our data the convulsant bemegride ranks first among non-antiepileptic candidates for epilepsy. We test two corrections for this confound, naive area-score subtraction and a negative-control method adapted from genomics and pharmacoepidemiology, across five indications. Naive wins every bias-diagnostic metric measured. Fixing a bug that silently zeroed the principled method's penalty term for held-out compounds changes the picture: it stays modest for depression, antipsychotic, and lipid, matching a direct intervention test that dials each correction's real effect into the score and finds the bias metrics genuinely unmoved by the principled method there; for epilepsy, our own motivating case, it does not, and a smaller but real response appears under the same intervention. A controlled simulation, 30 independent runs, explains both outcomes with one mechanism: the principled method's factor recovers the true confound better than naive's own penalty does (0.84 vs. 0.68 AUROC against ground truth), yet projecting it out of the final score, the method's actual correction step, fails to reduce confounding in the ranked list; subtracting that same estimated factor directly, instead of projecting it out, does (paired gap 0.22, $d_z = 3.0$, $p < 10^{-15}$), a gap in application mechanism, not in what was identified. The method identifies the confound; the specific way it is applied does not act on what it identifies, a distinct and more general finding than metric blindness.

1 Introduction

A bias-diagnostic metric is itself a measurement, and a measurement has a detection limit. When two debiasing corrections are compared against one bias metric

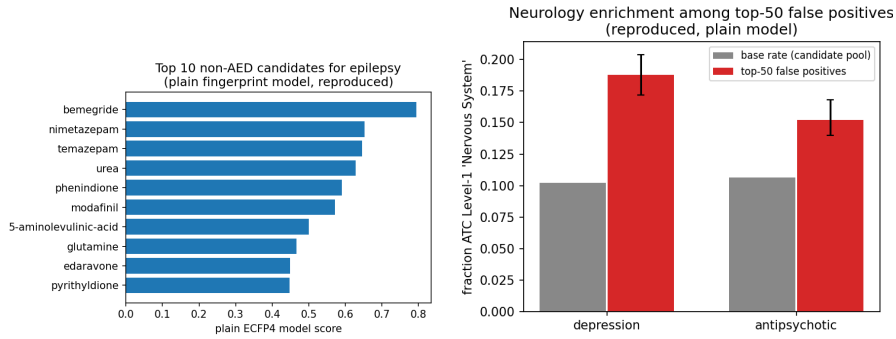


Fig. 1. Left: top 10 non-antiepileptic candidates ranked by a plain ECFP4 model; bemegride, a convulsant, ranks first. Right: nervous-system drugs (ATC 'N') enriched among top-50 false positives of depression/antipsychotic models vs. base rate.

and one wins, the usual reading is that the winning correction removes the bias better. A second reading is available whenever the two work through different mechanisms: the metric may simply be built to see one and not the other, in which case the comparison never tested what it looked like it was testing. Telling these readings apart requires checking, directly, whether the metric could have detected the losing correction's effect at all, rather than inferring quality from the scoreboard alone. We call this an alignment check: score each correction's own penalty term, the quantity it subtracts or projects out, as a standalone classifier of whatever the metric flags. A term that classifies the flag near chance is a screening signal, not proof, that the metric cannot reward or penalize that mechanism; Section 5.6 tests this directly.

We develop this check through a case study chosen because it produces exactly the puzzle the check is built to resolve. Chemical structure is not the only thing a repurposing model can learn to exploit: a model trained to separate antiepileptic drugs from everything else also, incidentally, learns what distinguishes neurology drugs from the rest of the pharmacopeia, and that second signal rides along with the first. In a plain ECFP4 fingerprint model trained on known antiepileptic drugs, bemegride, a convulsant once used in electroconvulsive therapy, ranks first among 1,522 non-antiepileptic candidates. We reproduce this exactly (score 0.7942) as this paper's motivating case. A companion study from this group audited evaluation-protocol choices in LINCS-based repurposing models and reported this same ranking [12], proposing no correction method.

The same pattern appears in aggregate. Restricting to drugs with a WHO ATC classification, nervous-system drugs make up 18.8% of the top 50 false positives for a depression model and 15.2% for an antipsychotic model, against base rates of 10.2% and 10.6%, 1.5 to 1.8 times base rate, smaller than a name-matched Hub label had suggested; ATC covers only 9.7% of the LINCS library.

Two corrections suggest themselves. The simple one subtracts a generic area score from the disease-specific score, directly penalizing confounded-class membership; prior informal analysis assumed this collapses precision too far to use. The more principled one borrows an idea from genomics and pharmacoepidemi-

ology: estimate the shared organ-system factor from negative-control indications, same area, different mechanism, and remove only that factor rather than a blunt membership score. Built and evaluated, it did not beat the simple one on any bias-diagnostic metric measured, where this paper could have stopped and reported a negative result.

Instead we apply the alignment check, and it complicates the conclusion rather than resolving it. Section 5.5 shows the bias metrics' flags are predictable from the same base representation for three of five targets but not uniformly, and a direct intervention test (Section 5.6) confirms the metric is genuinely insensitive to the principled method there, with a real, smaller exception for epilepsy. A synthetic experiment with a known confound (Section 5.6) explains why: the method's own factor recovers the true confound better than naive's penalty does, but how it is applied, fit jointly then projected out at scoring, does not reduce the confound's presence in the ranking the way naive's cruder subtraction does.

2 Related Work

Negative controls for unwanted variation. RUV-2 [8] estimates latent confounding factors from known-null control genes and removes them before testing; later work [9] unifies several RUV variants. Surrogate variable analysis [16] and empirical-Bayes batch correction [11] address related but distinct versions; ComBat needs known batch labels, not negative controls. Negative-control exposures in pharmacoepidemiology [18] formalize the same logic for drug confounding [23]. Our area-factor step is a direct structural analogue, indications rather than genes.

Debiasing drug-repurposing and DTI models. NRBdMF [1] factorizes a signed drug×disease matrix from SIDER [14] to reduce false positives through within-drug directionality; it never uses other diseases as negative controls. TAPB [17] and DRDM [19] address other biases (target-popularity, graph-degree); neither targets organ-system confounding. Campillos et al. [3,13] first showed SIDER-derived phenotypic similarity carries repositioning signal, the basis for our hard negatives.

Benchmarks and resources. repoDB [2] provides a curated gold-standard set of approved/failed drug-disease pairs, a standard secondary comparison point. The Drug Repurposing Hub [5], a name-matched label used in prior work, has limited coverage; we replace it with ATC classification. Epileptogenic/proconvulsant lists have an established clinical literature base [20,4,21]. On evaluation validity: scaffold splits overestimate performance relative to a realistic split [10], and affinity benchmarks reward memorization unless reweighted [25]; neither asks whether a bias-diagnostic metric can see a correction's mechanism at all.

3 Data

We build on the nine ChEMBL [24] indication classes used in our companion study (full list in the repository), over the same 9,878-compound LINCS library, extended with nine further ATC-defined indications so every original class has at least one same-area, different-mechanism negative-control partner. ATC Level-1 codes come from the ChEMBL molecule API by full InChIKey; only 954 of 9,878 compounds (9.7%) carry any ATC classification, why Section 5.1's enrichment is smaller than a name-matched Hub label had suggested.

SIDER [14] side-effect data is joined by PubChem CID, extending a prior 1,725-compound join; coverage remains low, 543 of 9,878 (5.5%), for the same reason as ATC. We use its adverse-effect terms as hard negatives (convulsion-family for epilepsy, hypotension for antihypertensives, hyperglycaemia for antidiabetics) where such a term map makes sense, not invented otherwise.

For epilepsy, we use a 43-drug proconvulsant list from this group's own prior, unpublished work, stated plainly rather than as an external resource: literature-curated from Chen, Albertson & Olson [4], a peer-reviewed review of drug-induced seizures, cross-validated against two public sources. Of 82 canonical compounds, 62 resolve to our library; 38 (61%) carry a SIDER-listed convulsion/seizure effect, and 24 (39%) match a risk tier in Ruffmann, Bogliun & Beghi's independent review [20].

4 Method

Each compound is represented by concatenating a 32-component PCA of its ECFP4 fingerprint (TruncatedSVD, no indication-matrix leakage) with a 16-dim embedding from a free logistic matrix factorization fit jointly across all 18 indications, its gradient masked to training cells only within each fold, so held-out labels never enter the embedding. Chemistry is necessary, not cosmetic: a pure co-occurrence factorization carries almost no signal here, 94% of the library has zero positive indications, why NRLMF/NRBdMF add drug or disease similarity; we add chemistry for the same reason. Under CVS2 a held-out compound's row is masked for every column, so its MF embedding stays at random initialization, injecting noise into the logit at roughly 5–10% chemistry's magnitude. Dropping the MF embedding and rerunning CVS2 on chemistry alone (code/24_chemistry_only_ablation.py) reproduces Tables 1–2 within 0.02 and epilepsy's Table 3 row within 0.001; depression's and antipsychotic's rows shift up 0.03–0.07 without this noise but stay well below naive's 0.60–0.79, so no claim below depends on it.

On top of this representation we compare four scorings for target indication k_0. *Plain* fits one logistic regression per indication, no area correction. *Naive subtraction* scores the plain score minus λ times a min-max-normalized mean indication rate over k_0's negative-control set, $\lambda = 2$. *NRBdMF* factorizes a signed drug-by-disease matrix [1]: -1 ChEMBL/ATC indication, $+1$ SIDER

adverse-effect hit, chemistry withheld, matching the original specification. *Proposed*: for k_0's negative-control set $\mathrm{NC}(k_0)$, same ATC area, different mechanism, take the train-only residuals of the already-fit plain models on $\mathrm{NC}(k_0)$'s columns, their leading left singular vectors as area factor A (Gagnon-Bartsch and Speed's RUV-2 strategy [8], applied to indications rather than genes), refit k_0 with [chemistry+MF, A], score with A's coefficients dropped.

We later tested a shrunk version of this correction, scaling A by $s \in [0, 1]$ following Gerard and Stephens' unified RUV family [9]; Section 5.4 reports no improvement.

4.1 The alignment check

Any correction scoring a candidate as a base score minus (or plus) a penalty term, naive subtraction's λ-weighted area rate and the proposed method's projected-out A-coefficients both have this form, can be probed independently of whatever metric judges it. Three scores matter for the proposed method: $s_{\text{plain}} = \hat{\beta}_0^\top x$ (fit without A), $s_{\text{joint}} = \hat{\beta}_X^\top x + \hat{\gamma}^\top a$ (refit with A), and $s_{\text{proposed}} = \hat{\beta}_X^\top x$ (A dropped). $P = s_{\text{joint}} - s_{\text{proposed}} = \hat{\gamma}^\top a$ is what this check and Section 5.6 score, not $s_{\text{plain}} - s_{\text{proposed}}$, since refitting with A also moves the base coefficients ($\mathrm{corr}(s_{\text{plain}}, \sigma(\hat{\beta}_X^\top x)) = 0.96$–$1.00$, coefficient shift 2–12% across targets, code/19_intervention_doseresponse.py; close here, not identical, and their top-50 rankings overlap 84–98% with rank correlation 0.98–1.00, narrowest for epilepsy, the target whose base coefficients shift most). Given a bias-diagnostic metric defined through a binary flag F (same-area membership, hard-negative membership, or any other criterion the metric penalizes), the check scores the correction's own penalty term P as a standalone classifier of F via $\mathrm{AUROC}(P, F)$. Near-0.5 AUROC means P carries little marginal rank information about F in isolation: a heuristic screening diagnostic, not a formal necessary condition: an interaction with the base score could move a ranking metric without a marginal association. Section 5.6 tests that directly; short of it, a comparison against a baseline whose term *is* informative about F is not on equal footing. We apply this to our two corrections in Section 5.5.

Section 5.5's result has a structural reason. **Proposition.** Let X be the shared representation on which every negative-control indication's model is fit by L_2-penalized logistic regression at inverse-strength C (weights W), coefficients $\hat{\beta}_j$, fitted probabilities $\hat{P}$, train rows only. Each residual $r_j = y_j - \hat{p}_j$ satisfies $X_{\text{tr}}^\top W r_j = \hat{\beta}_j / C$: orthogonality to X is exact only as $C \to \infty$; at finite C the deviation is proportional to $\hat{\beta}_j$, not zero. A, the leading left singular vectors of $R = [r_1, \ldots, r_{|\mathrm{NC}|}]$, inherits whatever orthogonality R has.

We measured this rather than assuming it away: at the $C = 1$ used throughout, A's projection onto X's column space holds 34–40% of A's variance, not the near-zero a pure guarantee would give; Section 5.5's unregularized OLS fit is where the exact case holds. Section 5.5 tests both halves empirically.

Table 1. AUPRC on true indications, CVS2 (mean over 5 reruns; 95% bootstrap CI in parentheses).

Target	Plain	Naive subtract	NRBdMF	Proposed
depression	0.201 (.114,.289)	0.190 (.105,.262)	0.039	0.181 (.112,.248)
antipsychotic	0.440 (.252,.628)	0.412 (.215,.608)	0.039	0.438 (.264,.612)
epilepsy	0.091 (.065,.127)	0.090 (.067,.128)	0.024	0.085 (.065,.112)

Table 2. Bias-diagnostic metrics. Top-50 same-area rate: lower is better (base rate ≈ 0.09–0.11). Hard-negative/proconvulsant rank: higher percentile is better.

Metric / Target	CVS	Plain	Naive subtract	NRBdMF	Proposed
area rate, depression	2	0.112	0.024	0.084	0.112
area rate, antipsychotic	2	0.136	0.096	0.084	0.144
area rate, epilepsy	2	0.144	0.068	0.120	0.180
hard-neg. pct., epilepsy	2	38.4	50.9	47.8	41.9
proconv. pct., epilepsy	2	30.9	71.8	49.2	26.4

5 Experiments

5.1 Reproducing the motivating confound

We retrained the plain ECFP4 model for epilepsy, depression, and antipsychotics, averaged over five seeds, confirming both parts of Section 1's motivating observation exactly: bemegride ranks first of 1,522 non-antiepileptic candidates, and the same 1.5–1.8x nervous-system enrichment reproduces for depression and antipsychotic. We could not locate, in any saved script, the AUPRC collapse from 0.45 to 0.17 an earlier analysis attributed to naive subtraction; we report what we can measure.

5.2 Evaluation protocol

We evaluate under three cross-validation schemes: CVS1 (random 20% cell hold-out), CVS2 (cold-start-drug, 20% of compounds), CVS3 (cold-start-disease, the entire target column), each run five times on an identical 1,702-compound pool shared by all four methods.

5.3 Main results

Table 1 reports CVS2 AUPRC on the true indication label (CVS1 and CVS3 tables, same pattern, are in the repository); Table 2 reports the two bias-diagnostic metrics that matter most here, the same-ATC-area rate among the top 50 false positives and the median rank percentile of SIDER-flagged hard negatives.

NRBdMF performs close to chance everywhere (AUPRC 0.02 to 0.15): a pure within-drug signed factorization, no chemistry, has little to work with on an 18-indication matrix this sparse. The proposed method tracks plain closely on AUPRC, the precision-preservation property it was designed for, but on the depression/antipsychotic/epilepsy bias metrics in Table 2 it is dominated by naive

subtraction; the clearest case is the proconvulsant panel, where naive pushes it from the 31st percentile to the 72nd while proposed moves it under 5 points, and in the epilepsy CVS2 row in the wrong direction. Naive's AUPRC cost is modest across every row and CVS scheme, never approaching the collapse an earlier informal analysis had reported (Section 5.1). Betablocker shows no method differentiation (area rate 0.108 to 0.112); lipid reproduces the core pattern (naive halves it, 0.104 to 0.052, at no AUPRC cost; proposed moves neither). A reconstructed DRDM baseline behaves like plain throughout, consistent with targeting a different bias (Section 2).

5.4 Robustness, validity, and identifiability follow-ups

A method that identifies an area factor from negative controls should, if those controls are mechanistically independent, leave each control's predictability roughly unchanged when corrected using a factor from the *other* controls. This leave-one-negative-control-out diagnostic, every (target, control) pair: 13 of 14 cells cluster within ± 0.03 of zero AUPRC change, consistent with independence (the exception, anxiolytic, is small-sample noise from its 7-positive class: 20 seeds shrink its mean delta to $+0.034$, within 1 SE of zero). Two further candidates tested and set aside. First, that the negative-control classes are small, 4 to 21 positives each, so the area factor is mostly noise: empirical-Bayes-style shrinkage, $A \leftarrow sA$ for $s \in \{0, 0.25, 0.5, 0.75, 1.0\}$ selected via the same diagnostic, picked full-strength, unshrunk correction ($s = 1$) for all three targets. Second, that RUV's zero-loading assumption [9] fails, so the area factor absorbs target signal along with the confound: regressing the true label (train rows) onto the fitted area factor gives R^2 0.0006–0.021 for all six cells, not tracking naive's advantage (epilepsy has the smallest overlap but the largest gap in naive's favor).

5.5 Applying the alignment check

Finding no identifiability violation leaves a more mundane candidate: the bias metrics may sit closer to naive's raw negative-control membership than to proposed's residualized factor. We apply the alignment check (Section 4.1): for each target, under CVS2, compute naive's penalty term and proposed's term (the contribution A's coefficients would have made, had they not been dropped) for every non-indicated test compound, and score each as a classifier of the exact compounds the bias metrics flag: same-ATC-area, SIDER-hard-negative, or proconvulsant-panel membership.

An earlier version of this table scored test rows using the SVD's own output directly, exactly zero there by construction (a zero residual row forced by the negative controls' own train mask, $\sim 10^{-16}$): the original "0.50 AUROC" was the AUROC of a value mathematically zero for every test compound, not a property of the method. We instead fit a linear out-of-sample projection from X to the train rows' own factor values (5-fold CV R^2 0.44–0.56, non-degenerate) for test rows below.

Table 3. AUROC of each quantity vs. the flagged compounds (mean, 5 seeds, CVS2). 0.5 = no information. P and Plain→Proposed are sign-free PER SEED before averaging, not after (a bug in the latter, found this round, inflated depression's/antipsychotic's P toward chance; see text); Plain→Proposed is the full empirical plain-vs-proposed gap.

Target	Flag	Naive	P	Plain→Proposed	X alone
depression	same-ATC-area	0.766	0.561	0.575	0.718
depression	hard-negative	0.636	0.540	0.556	0.514
antipsychotic	same-ATC-area	0.774	0.579	0.568	0.706
antipsychotic	hard-negative	0.663	0.580	0.568	0.591
epilepsy	same-ATC-area	0.787	0.657	0.653	0.722
epilepsy	hard-negative	0.600	0.568	0.559	0.613
epilepsy	proconvulsant	0.721	0.676	0.677	0.695

Naive's term separates every flag well above chance, 0.60 to 0.79, unaffected by either fix. A decomposition this round's review asked for surfaced a second bug: aggregating by taking $\max(\text{AUROC}, 1 - \text{AUROC})$ of the AUROC already averaged across seeds, rather than of each seed's own AUROC before averaging, lets a real per-seed effect cancel toward 0.5 whenever the SVD factor's arbitrary sign lands differently across independent refits, exactly depression's and antipsychotic's cells (epilepsy's sign is stable, unaffected). Corrected to the per-seed convention code/23_permutation_check.py already used, P is 0.54–0.58 for depression and antipsychotic, modest rather than literally at chance, still well below naive's 0.64–0.77. Plain→Proposed, the full empirical plain-to-proposed gap, is comparable to P on most cells (0.53–0.68 both) but not identical, related not interchangeable (Section 4.1); against the undropped joint fit instead of proposed, that same gap is lower on 6 of 7 cells (results/score_decomposition_alignment.csv), the two effects partially cancelling rather than adding when combined into one score. $\max(\text{AUROC}, 1 - \text{AUROC})$ is itself upward-biased under the null when its sign is picked on the same evaluation set it is reported on. A permutation test (shuffle the flag within each seed's split, 5000 times, code/23_permutation_check.py), combined across seeds with Fisher's method against a Bonferroni threshold for the 7 cells ($\alpha = 0.0071$): depression stays non-significant ($p = 0.12, 0.66$); antipsychotic's two flags and epilepsy's hard-negative pass on all 5 seeds ($p = 0.0028$–0.024) but collapse ($p \geq 0.18$) with the single most extreme seed dropped, a seed-specific artifact; epilepsy's same-ATC-area holds up either way ($p = 2.2 \times 10^{-9}$ all 5 seeds, $p = 5.4 \times 10^{-7}$ excluding the most extreme). Proconvulsant clears the raw threshold ($p = 0.0004$) but not α once that seed is dropped ($p = 0.01$). Fisher's method also assumes the 5 seeds are independent; they are not, since all 5 resplit one compound pool. A compound-level permutation modeling that directly (code/25_hierarchical_permutation.py) agrees with Fisher except on proconvulsant, where it finds significance ($p = 0.0027$) the leave-one-out check does not: real under the test built for this concern, fragile under the other. By the Proposition, proposed's term is only partially orthogonal, on train rows, to X; the rightmost column tests, rather than assumes, whether F is predictable from X. Same-ATC-area is well predicted by X on every target, 0.706 to 0.722, close to naive's own 0.766 to 0.787: most of what marks a compound as belonging to the confounded area is already present in X before any negative-control informa-

tion is added. This does not explain why epilepsy's area term specifically carries signal while depression's and antipsychotic's do not; Section 5.6 does.

This two-step recipe names nothing drug-repurposing-specific; it diagnoses association, not what the metric does when the real effect is applied, which Section 5.6 tests directly.

We also re-ran this check outside drug repurposing, on two genomics datasets with a known ground-truth confound: Vawter et al.'s brain gene-expression study [22] (gender vs. lab, 799 housekeeping controls [7]) and bladderbatch [6,15] (3-level cancer status vs. 5-level batch). Step one holds exactly in both: each control gene's OLS residual satisfies the normal equations to floating-point precision. Step two diverges: lab is unpredictable from gender (AUROC 0.39–0.41) and the orthogonalized factor stays informative about it (0.50–0.94 alone, 0.95–1.00 combined, beating naive's 0.55–0.76); batch is moderately predictable from cancer status (0.722), yet the orthogonalized factor stays as informative about batch as naive's own (0.68 vs. 0.62). Predictability of F from X alone, not whether F and the covariate of interest are related, is what the check's second step turns on.

5.6 Confirming it directly: intervention, and a synthetic ground truth

Standalone AUROC does not show whether Table 2's metric actually moves when a penalty's real effect is dosed into the score. We sweep how much of each penalty is included, $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ ($\alpha = 1$ for naive and $\alpha = 0$ for proposed reproduce Table 2), and recompute the bias metrics at each dose, same CVS2 split, all five targets. Naive's metrics move strongly and monotonically for depression, antipsychotic, and lipid (epilepsy proconvulsant percentile: 30.9 at $\alpha = 0$ to 71.8 at $\alpha = 1$); proposed's stay exactly flat for the same three (depression area rate 0.112, lipid 0.104, throughout). Betablocker moves little under either. Epilepsy again exceptions: proposed's metrics move a small real amount (area rate 0.180 to 0.140, proconvulsant percentile 26.4 to 32.7), a quarter naive's swing. This confirms the alignment claim operationally for three of five targets and the same exception as Table 3; unlike standalone AUROC, this measurement is not subject to Section 5.5's sign-selection caveat, so epilepsy's small shift stands as independent evidence regardless.

A real-data flag is a proxy; the true confound is unobserved. A simulation fixes that: 3,000 compounds, chemistry X, an organ-area confound z noisily tied to a chemistry subspace, indication log-odds combining a disease-specific term with, for same-area diseases, z; negative controls share the area with an independently drawn mechanism, over 30 independent seeds, each redrawing X, loadings, and labels. AUPRC (0.484/0.273/0.485 plain/naive/proposed) reproduces Table 1's pattern. Figure 2a compares five scores on the top-50 rate of *true* high-confound compounds (base rate 0.29): naive substantially reduces it, proposed does not; direct subtraction of proposed's own estimated factor, the identical estimate applied the other way, reduces it as much as naive does, a paired gap of 0.218 (95% CI 0.191–0.245, $d_z = 3.0$, $p < 10^{-15}$) attributable to

application mechanism alone. Proposed's factor AUROC against the true confound is 0.84, naive's own 0.68: it identifies the confound better, but projecting it out at scoring does not act on what it identified. Figure 2b sweeps estimation quality directly, the same two mechanisms on the same noisy estimate at seven levels: the gap is near zero at the noise-free extreme, peaks at moderate noise near the method's own 0.84-AUROC quality, and narrows again at high noise as both converge to the uncorrected baseline, non-monotonic rather than growing without bound. Where the factor carries little real signal (depression, antipsychotic, lipid) there is little for this gap to act on; where it does (epilepsy), the gap becomes the effect measured here.

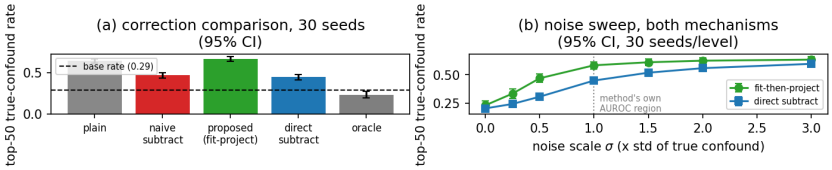


Fig. 2. Synthetic experiment, 30 seeds, 95% CI. (a) Top-50 true-confound rate by score. (b) Both application mechanisms over seven noise levels on the identical estimate.

This gap holds across a tenfold range of confound-chemistry correlation (code/28_confound_chemistry_correlation_sensitivity.py, $R^2(z, X)$ 0.07–0.80, paired gap 0.03–0.26), vanishing only where X already explains z almost entirely ($R^2 \geq 0.94$): specific to confounds the representation does not already absorb, not an artifact of one fixed structural choice.

6 Discussion and Limitations

This paper set out to build a more principled alternative to a correction assumed too costly to use; naive won every bias metric measured. Whether that scoreboard meant what it appeared to is target-dependent: for depression, antipsychotic, and lipid the metric genuinely cannot see the proposed method's mechanism; for epilepsy's area and proconvulsant flags, it partially can (Section 5.6). Winning a bias metric means suppressing a flagged signal, not a validated improvement: organ-system overlap can itself be real pharmacology, so naive's larger reduction is not shown to yield better candidates, only a stronger diagnostic response.

The check and the correction-application gap are the more durable contributions; the residual-orthogonality result (Proposition, Section 4.1) follows from standard regression score equations, so the contribution is the diagnostic framework built on it and the experiments confirming it applies, not new theory. A caveat: a bug zeroed the correction's out-of-sample factor for every held-out compound until this revision (Section 5.5), producing a cleaner "0.50 everywhere" result than the method earns. DRDM [19] has no public code; its baseline is a reconstruction, not a verified reproduction.

Limitations: ATC/SIDER cover under 10% of the library; CVS3 collapses plain and proposed together; repoDB [2] gives too little overlap for an AUPRC check. Code: `https://github.com/edanmn/area-debiasing-drug-repurposing`.

References

1. Azuma, I., Mizuno, T., Kusuhara, H.: NRBdMF: A recommendation algorithm for predicting drug effects considering directionality. Journal of Chemical Information and Modeling **63**(2), 474–483 (2023). https://doi.org/10.1021/acs.jcim.2c01210
2. Brown, A.S., Patel, C.J.: A standard database for drug repositioning. Scientific Data **4**, 170029 (2017). https://doi.org/10.1038/sdata.2017.29
3. Campillos, M., Kuhn, M., Gavin, A.C., Jensen, L.J., Bork, P.: Drug target identification using side-effect similarity. Science **321**(5886), 263–266 (2008). https://doi.org/10.1126/science.1158140
4. Chen, H.Y., Albertson, T.E., Olson, K.R.: Treatment of drug-induced seizures. British Journal of Clinical Pharmacology **81**(3), 412–419 (2016)
5. Corsello, S.M., Bittker, J.A., Liu, Z., Gould, J., McCarren, P., Hirschman, J.E., Johnston, S.E., Vrcic, A., Wong, B., Khan, M., Asiedu, J., Narayan, R., Mader, C.C., Subramanian, A., Golub, T.R.: The drug repurposing hub: A next-generation drug library and information resource. Nature Medicine **23**(4), 405–408 (2017). https://doi.org/10.1038/nm.4306
6. Dyrskjot, L., Kruhøffer, M., Thykjaer, T., Marcussen, N., Jensen, J.L., Møller, K., Ørntoft, T.F.: Gene expression in the urinary bladder: A common carcinoma in situ gene expression signature exists disregarding histopathological classification. Cancer Research **64**(11), 4040–4048 (2004)
7. Eisenberg, E., Levanon, E.Y.: Human housekeeping genes are compact. Trends in Genetics **19**(7), 362–365 (2003)
8. Gagnon-Bartsch, J.A., Speed, T.P.: Using control genes to correct for unwanted variation in microarray data. Biostatistics **13**(3), 539–552 (2012). https://doi.org/10.1093/biostatistics/kxr034
9. Gerard, D., Stephens, M.: Unifying and generalizing methods for removing unwanted variation based on negative controls. Statistica Sinica **31**, 1145–1166 (2021)
10. Guo, Q., Hernandez-Hernandez, S., Ballester, P.J.: Scaffold splits overestimate virtual screening performance (2024)
11. Johnson, W.E., Li, C., Rabinovic, A.: Adjusting batch effects in microarray expression data using empirical bayes methods. Biostatistics **8**(1), 118–127 (2007). https://doi.org/10.1093/biostatistics/kxj037
12. Kondadadi, R.K., McCreary, D.: How much do evaluation choices inflate transcriptomic drug repurposing? a measured audit on LINCS L1000 (2026), manuscript in preparation
13. Kuhn, M., Campillos, M., Letunic, I., Jensen, L.J., Bork, P.: A side effect resource to capture phenotypic effects of drugs. Molecular Systems Biology **6**, 343 (2010). https://doi.org/10.1038/msb.2009.98
14. Kuhn, M., Letunic, I., Jensen, L.J., Bork, P.: The SIDER database of drugs and side effects. Nucleic Acids Research **44**(D1), D1075–D1079 (2016). https://doi.org/10.1093/nar/gkv1075
15. Leek, J.T., Scharpf, R.B., Corrada Bravo, H., Simcha, D., Langmead, B., Johnson, W.E., Geman, D., Baggerly, K., Irizarry, R.A.: Tackling the widespread and critical impact of batch effects in high-throughput data. Nature Reviews Genetics **11**, 733–739 (2010)
16. Leek, J.T., Storey, J.D.: Capturing heterogeneity in gene expression studies by surrogate variable analysis. PLoS Genetics **3**(9), e161 (2007). https://doi.org/10.1371/journal.pgen.0030161

17. Lin, G., Zhang, X., Ren, Z., Zou, Q., Tiwari, P., Zhou, C., Ding, Y.: TAPB: An interventional debiasing framework for alleviating target prior bias in drug-target interaction prediction. Nature Communications **16** (2025). https://doi.org/10.1038/s41467-025-66915-1
18. Lipsitch, M., Tchetgen Tchetgen, E., Cohen, T.: Negative controls: A tool for detecting confounding and bias in observational studies. Epidemiology **21**(3), 383–388 (2010). https://doi.org/10.1097/EDE.0b013e3181d61eeb
19. Meng, Y., Wang, Y., Hu, X., Lu, C., Tang, X., Cui, F., Zeng, P., Yao, Y., Yang, J., Xu, J.: Adaptive debiasing learning for drug repositioning. Journal of Biomedical Informatics **167**, 104843 (2025). https://doi.org/10.1016/j.jbi.2025.104843
20. Ruffmann, C., Bogliun, G., Beghi, E.: Epileptogenic drugs: A systematic review. Expert Review of Neurotherapeutics **6**(4), 575–589 (2006)
21. Shorvon, S., Tallis, R., Wallace, H.: Antiepileptic drugs: Coprescription of proconvulsant drugs and oral contraceptives: A national study of antiepileptic drug prescribing practice. Journal of Neurology, Neurosurgery & Psychiatry **72**, 114–115 (2002)
22. Vawter, M.P., Evans, S., Choudary, P., Tomita, H., Meador-Woodruff, J., Molnar, M., Li, J., Lopez, J.F., Myers, R., Cox, D., Watson, S.J., Akil, H., Jones, E.G., Bunney, W.E.: Gender-specific gene expression in post-mortem human brain: Localization to sex chromosomes. Neuropsychopharmacology **29**(2), 373–384 (2004)
23. Zafari, Z., Park, J.e., Shah, C.H., dosReis, S., Gorman, E.F., Hua, W., Ma, Y., Tian, F.: The state of use and utility of negative controls in pharmacoepidemiologic studies. American Journal of Epidemiology **193**(3), 426–453 (2023). https://doi.org/10.1093/aje/kwad201
24. Zdrazil, B., Felix, E., Hunter, F., Manners, E.J., Blackshaw, J., Corbett, S., de Veij, M., Ioannidis, H., Lopez, D.M., Mosquera, J.F., Magarinos, M.P., Bosc, N., Arcila, R., Kizilören, T., Gaulton, A., Bento, A.P., Adasme, M.F., Monecke, P., Landrum, G.A., Leach, A.R.: The ChEMBL database in 2023: A drug discovery platform spanning multiple bioactivity data types and time periods. Nucleic Acids Research **52**(D1), D1180–D1192 (2024). https://doi.org/10.1093/nar/gkad1004
25. Zhang, C., Hu, Z., Jiang, C., Chen, W., Xu, J., Zhang, S.: Rethinking the generalization of drug-target affinity prediction algorithms via similarity-aware evaluation. In: International Conference on Learning Representations (ICLR) (2025)